Applied Economics

Causality and Correlation

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Plan for this Lecture

1. Correlation vs. Causality
2. Endogeneity
    - Reverse Causality
    - Measurement Error
    - Omitted Variable Bias
3. Model Specification

Correlation vs. Causality

Correlation vs Causation

Correlation is a **statistical measure** describing the **relation** between treatment variable x and outcome variable y.

"Those who have a university degree earn more that those who do not"

"When icecream consumption increases, sea drownings increase"

"Healthier people are more wealthy"

Causation is a **claim** affirming that treatment x causes outcome y.

"Having a university degree makes you earn more" Selection Bias

"Icecream consumption makes pepole drown in the sea" Ommitted Variable

"Better health makes people more wealthy" Reverse Causality

A correlation **does not necessarily** reflect a causal relationship. Why?

Endogeneity: Selection bias, omitted variables, reverse causality...

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Causal Inference

Causal inference consists on using econometric tools to **credibly** support causal claims.

1. First-best: field experiments or **randomised control trials (RCTs)**
   - A "treatment" is randomly allocated to find its causal effect on an outcome.
   - Compare mean outcome of treatment and control groups to infer causal effect.
   - Popular in development economics, but often unethical or impractical in other fields.
2. Second-best: quasi-experiments or natural experiments.
   - Setups where external events, policies, or random conditions created a random allocation of the treatment.
3. Third best: non-experiments, observational data.
   - Non-random allocation of treatment. Claiming causality becomes tricky.
   - Use multiple linear regression (MLR) to reduce endogeneity concerns.

This course will focus on MLR and quasi-experimental methods.

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Big Picture: Steps in empirical economic analysis

Economic question: does education cause an increase in wages?

1. Construct an economic model for the representative agent.

$$wage = \beta_0 + \beta_1 educ + \beta_2 expert + \beta_3 ability$$

2. Turn the economic model into a statistical model for the population.

$$wage_i = \beta_0 + \beta_1 educ_i + \beta_2 expert_i + \beta_3 ability_i + u_i \quad i = 1, ..., N$$

3. Turn the statistical model for the observed sample.

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When we change the value of X **without changing anything else**, then Y would also change as a result.

An ideal way to identify causal effects are **Randomized Control Trials** (RCTs)

- you randomly assign people to a treatment and a control group.
  - $\rightarrow$ the treatment assignment is independent of all other potential factors that have an impact on Y.
  - $\rightarrow$ treatment effects can be estimated by simply comparing means across groups
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Causal Analysis with Observational Data

RCT disadvantages

- Expensive.
- Ethical issues.
- External validity issues: to what extent results from a (small) RCT can be generalized to other contexts (e.g. other countries).

Thus, we often use observational data (as opposed to experimental data).
With this type of data, causal inference is usually more difficult.

Coefficients in a (multiple) linear regressions measure (partial) correlations between variables. They can, but they don't have to reflect causal effects!

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Exogeneity and Endogeneity

Endogeneity

Exogeneity (Assumption 4) is central to obtain causal effects.

$$E(u_i|\boldsymbol{X}_i) = E(u_i) = 0$$

- Regressors $\boldsymbol{X}_i$ and the error term u_i are independent.
- The regressors $\boldsymbol{X}_i$ behave as if they had been randomly assigned.

Endogeneity $E(u_i|\boldsymbol{X}_i) \neq 0$

Correlation between one or more regressors $\boldsymbol{X}_i$ and the error term u_i.
Regressors $\boldsymbol{X}_i$ are not as if randomly assigned.

Implication: OLS estimator is biased and inconsistent:

$$E[\hat{\beta}] = \beta + E[(\boldsymbol{X}'\boldsymbol{X})^{-1}(\boldsymbol{X}'\boldsymbol{u})] \neq \beta$$

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Endogeneity Example in a Generic Simple Regression Model

$$Y_{it} = \beta_0 + \beta_1 X_{it} + u_{it} \quad i = 1, ..., N$$

Parameter estimate for the observed sample is:

$$\hat{\beta}_1 = \beta_1 + \frac{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}) \cdot u_i}{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

Under Assumptions 1 to 3:

- $\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}) \cdot u_i \xrightarrow{p} \text{cov}(X_i, u_i)$
- $\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 \xrightarrow{p} \text{var}(X_i)$

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\text{cov}(X, u)}{\text{var}(X)}$$

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Sources of Endogeneity

Frequent reasons for endogeneity are

- Omitted variables
- Measurement error in the explanatory variable
- Reverse causality.

Omitted Variables

Omitted Variables Example

Relevant model: $wage_i = \beta_0 + \beta_1\, educ_i + \beta_2\, ability_i + u_i$

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Estimated model: $wage_i = \beta_0 + \beta_1 educ_i + \beta_2 ability_i + v_i \quad v_i = \beta_2 \cdot ability_i + u_i$

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Endogeneity requires:

- $\beta_2 \neq 0$: omitted variable to correlate with outcome.
- $E[educ_i \cdot ability_i] \neq 0$: omitted variable to correlate with the regressor.

$$E[educ_i \cdot v_i] = \beta_2 \cdot E[educ_i \cdot ability_i] + \underbrace{E[educ_i \cdot u_i]}_{0}$$

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Direction of the Bias

Simmilarly, the covariance between error term and regresson can be expressed as:

$$\text{Cov}(X, u) \rightarrow \text{Cov}(\textit{educ}, v) = \beta_2 \,\text{Cov}(\textit{educ}, \textit{ability}) + \underbrace{\text{Cov}(\textit{educ}, u)}_{\rightarrow 0}$$

$$\hat{\beta}_1 \xrightarrow{p} \beta_1 + \frac{\text{Cov}(X, u)}{\text{Var}(X)} \quad \equiv \quad \beta_1 + \beta_2 \cdot \frac{\text{Cov}(\textit{educ}, \textit{ability})}{\text{Var}(\textit{educ})}$$

Direction of the **Omitted Variable Bias** depends on:

- the sign of β_2 (impact of ability on wages).

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- the direction of $\text{Cov}(\textit{educ}, \textit{ability})$ (the impact of ability on education)

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Omitted Variables: Example

In the Esitmated Model, $\hat{\beta}_1$ is **biased upward** because

- $\beta_2 > 0$: ability increases wages.
- $cov(IQ_i, educ_i) > 0$: *education* is positively correlated with the omitted variable

Thus, we *overestimate* the true positive impact of education on wages; part of the correlation reflects the *indirect* effect that individuals with higher education have a higher IQ, and therefore have higher wages.

In real-world applications, we do not have data on the omitted variable.
We can only speculate about its correlation with the included regressor(s) and its impact on the dependent variable, to think about how its omission affects the parameter of interest.

Measurement Error

(Classical) Measurement Error in the Regressor

Original model: $\text{wage}_i = \beta_0 + \beta_1 \text{educ}_i + u_i \quad E[u_i \mid \text{educ}_i] = 0$

We do not observe educ_i but just a noisy measure $\widetilde{\text{educ}}_i$: $\widetilde{\text{educ}}_i = \text{educ}_i + w_i$

- $E[w_i] = 0$
- $E[w_i \text{educ}_i] = 0$
- $E[w_i u_i] = 0$

Estimated model: $\text{wage}_i = \beta_0 + \beta_1 \widetilde{\text{educ}}_i + e_i \qquad e_i = -\beta_1 w_i + u_i$

$$
\begin{aligned}
Cov(X, u) = Cov(\widetilde{\text{educ}}_i, e_i) &= E(e_i \widetilde{\text{educ}}_i) \\
&= E[(-\beta_1 w_i + u_i)\widetilde{\text{educ}}_i] \\
&= -\beta_1 E[w_i \widetilde{\text{educ}}_i] + E[u_i \widetilde{\text{educ}}_i] \\
&= -\beta_1 E[w_i \text{educ}_i + w_i^2] + E[u_i \text{educ}_i + u_i w_i] \\
&= -\beta_1 E[w_i^2] \neq 0
\end{aligned}
$$

(Classical) Measurement Error in the Regressor

Original model: $\text{wage}_i = \beta_0 + \beta_1 \text{educ}_i + u_i \quad E[u_i \mid \text{educ}_i] = 0$

We do not observe educ_i but just a noisy measure $\widetilde{\text{educ}}_i$: $\widetilde{\text{educ}}_i = \text{educ}_i + w_i$

- $E[w_i] = 0$
- $E[w_i \text{educ}_i] = 0$
- $E[w_i u_i] = 0$

Estimated model: $\text{wage}_i = \beta_0 + \beta_1 \widetilde{\text{educ}}_i + e_i \qquad e_i = -\beta_1 w_i + u_i$

$$
\begin{aligned}
Cov(X, u) = Cov(\widetilde{\text{educ}}_i, e_i) &= E(e_i \widetilde{\text{educ}}_i) \\
&= E[(-\beta_1 w_i + u_i)\widetilde{\text{educ}}_i] \\
&= -\beta_1 E[w_i \widetilde{\text{educ}}_i] + E[u_i \widetilde{\text{educ}}_i] \\
&= -\beta_1 E[w_i \text{educ}_i + w_i^2] + E[u_i \text{educ}_i + u_i w_i] \\
&= -\beta_1 E[w_i^2] \neq 0
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Measurement Error causes attenuation bias

$$
\begin{aligned}
\hat{\beta}_1 \quad \xrightarrow{p} \quad & \beta_1 + \frac{Cov(\widetilde{\text{educ}}_i, e_i)}{Var(\widetilde{\text{educ}}_i)} \\
& = \beta_1 - \beta_1 \frac{E[w_i^2]}{Var(\text{educ}_i + w_i)} \\
& = \beta_1 - \beta_1 \frac{Var(w_i)}{Var(\text{educ}_i) + Var(w_i)} \\
& = \beta_1 \frac{Var(\text{educ}_i) + Var(w_i) - Var(w_i)}{Var(\text{educ}_i) + Var(w_i)} \\
& = \beta_1 \frac{Var(\text{educ}_i)}{Var(\text{educ}_i) + Var(w_i)}
\end{aligned}
$$

$0 < \frac{Var(\text{educ}_i)}{Var(\text{educ}_i) + Var(w_i)} \leq 1 \Rightarrow \hat{\beta}_1$ biased towards zero (attenuation bias)

Reverse Causality

Reverse Causality

We talk about reverse causality when Y causes X **and** X causes Y. **Previous Example:**

- We are interested in the effect of school budget on student outcomes:

$$grades_i = \beta_0 + \beta_1 funding_i + u_i$$

- But the grades also determine government spending on a school:

$$funding_i = \delta_0 + \delta_1 grades_i + v_i$$

- Assume: $cov(u_i, v_i) = 0$
- Still, we get endogeneity: $cov(funding_i, u_i) \neq 0$,
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Recap and Next Steps

Recap

- We are interested in estimating relationships between economic variables.
- This requires that we interpret the linear model as some stylized (economic) model, where the βs describe causal effects of $\boldsymbol{X}$ on Y.
- When we interpret coefficients in that way, the Exogeneity Assumption is crucial to get consistent estimates of the causal effects with OLS.
- OLS estimators $\hat{\beta}_j$ consistently estimate the **(partial) correlation** between X_j and Y. But if there is **endogeneity**, this does not represent the underlying causal relationship.
- Endogeneity can arise because of reverse causality, measurement error and omitted variables. It is an extremely frequent problem in the social sciences.

Next Steps

How can we address the endogeneity issue?

1. Instrumental Variables.
2. Panel Data.
3. Differences in Differences.
4. Regression Discontinuity Design.

APPENDIX

What do slope coefficients in a MLR measure? The Frisch-Waugh Theorem

In a MLR, $\hat{\beta}_j$ reflects the partial correlation between X_j and Y, after the effects of other regressors have been taken out.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \boldsymbol{X}'_{2i}\boldsymbol{\beta_2} + u_i, i = 1, \ldots, n$$

1. Regress X_{1i} on $\boldsymbol{X}_{2i}$: yields residuals $\hat{e}_{x_1 i}$
2. Regress Y_i on $\boldsymbol{X}_{2i}$: yields residuals $\hat{e}_{yi}$
3. Regress $\hat{e}_{yi}$ on $\hat{e}_{xi}$

$\rightarrow$ estimated slope parameter in step 3 is identical to the OLS estimator $\hat{\beta}_1$ in the MLR! (Frisch-Waugh Theorem)

The Frisch-Waugh Theorem

The error $\hat{e}_{x_1 i}$ can be thought of as the component (or variation) of X_{1i} that is not linearly explained by $\boldsymbol{X}_{2i}$.

Thus, the coefficient β_1 is a (rescaled) measure of linear correlation of X_{1i} and Y_i, after **partialling out (controlling linearly for) the effects of all variables in $\boldsymbol{X}_{2i}$**

$\rightarrow$ Basis for the **ceteris paribus interpretation** of the model coefficients in a multivariate regression set-up.

But: we keep only the other included regressors fixed. Causal inference requires that there are no components left in the error that are systematically correlated with X_1.